

Congruence certificates for Collatz-like machines: a census, a sieve completeness theorem, and the impossibility of separating two flagship orbits

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Abstract

We study *one-schema valuation machines*: Collatz-like functions on the positive integers determined by five integers $(\alpha, \beta, \gamma, \delta, \varepsilon)$, where for $x = 2^v u$ with u odd the function halts if $u = 1$ and otherwise, writing $u = 2k + 1$, returns $A_v k + B_v$ with $A_v = \alpha 2^{v+1} + \beta$ and $B_v = \gamma 2^v + \delta v + \varepsilon$. The class is of interest because the published halting-equivalent reduction of the *Space Needle*—a 6-state Turing machine and a well-known open problem in the busy-beaver community—is the member $(1, 3, 1, 1, 0)$.

We report four things. (i) A census of the 1,077 well-defined members of a parameter box, of which 318 are decided by an exact congruence procedure that is complete for its certificate class, with every certificate independently re-audited; and closed-form criteria for separation at moduli 3, 4, 6, 8, each verified exhaustively on $(\mathbb{Z}_m)^5$. (ii) A closed form for the last-step sieve of the family, from which ten members are proved never to halt after their first step, together with a *completeness theorem*: within the box those ten are all, every other applicable member carrying an explicit positive witness that the method fails on it. (iii) A chain of saturation theorems ending in: if β is odd, $A_v > 0$ and $\gcd(\delta, M') = 1$ (M' the odd part of M), then the congruence closure of every residue is all of $\mathbb{Z}_M$, at every modulus. Consequently *no congruence certificate at any modulus separates the Space Needle’s reduction*; the same argument transfers to base 3 and closes the corresponding question for a second machine studied here. (iv) A test of the theory outside its design set: a second cryptid found independently by the busy-beaver community (the “sheep” machine, April 2026) lies inside the interface, its last-step sieve closes completely—confirming a prediction made beforehand about which machines it should close for—and its halting set is obtained in closed form. (v) Expressiveness barriers for the class, and a refutation of counting-style arguments against its universality.

We are explicit about what is *not* achieved. No Turing machine’s halting problem is decided here; neither flagship orbit question is resolved; and every machine we do decide is, by the fact of admitting a certificate, not one of the hard ones. The principal results are impossibility theorems about a certificate class, and their value is explanatory. Each result in the paper carries an explicit epistemic label—proved, proved by exhaustive finite computation, machine-verified over a stated domain, or measured—and Section 9 tabulates all of them together with the threats to validity.

*This research program was carried out with an AI assistant (Claude, Anthropic). All proofs were additionally machine-checked as described in Section 1.5; the form of credit and the description of the workflow are to be settled before submission.

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1 Introduction

The $3x + 1$ problem asks whether a two-branch affine map sends every positive integer to 1; it has been open for close to a century [Lag85, Lag21]. Its modern relatives are *Collatz-like halting problems*: small, fully explicit piecewise-affine maps whose halting is equivalent to an orbit-avoidance question that resists every known technique. Conway showed that sufficiently general Collatz-like maps simulate arbitrary computation [Con72], and Kurtz and Simon placed the general halting problem for the class at Π_2^0 -complete [KS07]. Those reductions, however, require arbitrarily many independently chosen affine branches; they say nothing about an individual small map, nor about a parametrised family whose infinitely many branches are generated by a

fixed formula from finitely many integers. This paper studies one such family, chosen because the published halting-equivalent reduction of a well-known open Turing machine is a member of it.

1.1 Three kinds of object, and how they are related

Confusion between the following is easy and consequential, so we fix terminology first.

Turing machines. The busy-beaver community studies halting of concrete small Turing machines [bbcC]. The **Space Needle** is one: the 6-state machine

1RB1LA_1LCORE_1LF1LD_ORBOLA_1RC1RE_---OLD,

found by mx dys in January 2025 [bbcS]. Whether it halts is open.

Collatz-like functions. Certain Turing machines admit a *halting-equivalent reduction* to the orbit of an arithmetic map. For the Space Needle, Doucette’s one-variable reduction [bbcS] states: with $v(b)$ the 2-adic valuation of b and m its odd part, the machine halts if and only if the orbit of $b_0 = 6$ under

$$b \longmapsto b + v(b) + \frac{3}{2}(m - 1) \quad (\text{halting when } b \text{ is an exact power of two}) \quad (1)$$

reaches an exact power of two. A short computation identifies (1) as the member $(1, 3, 1, 1, 0)$ of Definition 2.1: writing $b = 2^v(2k + 1)$ gives $b + v + 3k = (2^{v+1} + 3)k + (2^v + v)$.

Counter and rewrite systems. A third genre—machines given by a handful of rewrite rules on tuples of counters—sits between the two. The second machine studied here, which we call **M3**, is of this kind: six rules on states $A(a, b)$, given in Section 6, whose halting reduces to a base-3 orbit question.

The consequence for reading this paper. Our census enumerates *Collatz-like functions*, not Turing machines: its members are five-tuples of integers and were never presented as machines in the busy-beaver sense. When we say a census member is “decided”, we mean a proof that the orbit of $x_0 = 3$ under that arithmetic map never reaches a power of two. No Turing machine’s halting problem is decided anywhere in this paper. The two directions in which our results do touch genuine machines are both negative: Corollary 5.6 and Theorem 6.4 say that a certain proof technique can never settle the Space Needle’s reduction, or M3.

1.2 The research goals

This paper reports part of a longer program. The program’s stated goals, in the order in which they compound, are:

- G1 (observe).** Establish new structural facts about Collatz-like functions: normal forms, invariants, closed-form criteria.
- G2 (decide).** Decide halting for specific machines wherever instance structure permits.
- G3 (explain).** Sharpen the account of *what makes the hard instances hard*—ideally by proving that named techniques cannot work, rather than by reporting that they have not worked.

G4 (transfer). Grow a toolkit that moves between machines rather than being rebuilt per instance.

1.3 Progress against the goals: summary

Section 10 gives the assessment in full, with what did not move. In brief:

goal	verdict	what this paper contributes
G1	substantial	the sieve lemma (Lemma 4.1) and its linear corollary; the completeness theorem (Thm 4.4); closed-form separation criteria at $m = 3, 4, 6, 8$ (Thm 3.4); the sieve-group rank dichotomy (Remark 6.5), now tested on a machine we did not construct (Remark 7.4)
G2	moved, but not on the target class	318 census members decided by certificate, 10 more by the sieve—all of them Collatz-like <i>functions</i> , none of them Turing machines, and none of them hard: a member with a certificate is by definition not a hard instance. Neither flagship orbit question is resolved. What is new here is a <i>theorem about a wild cryptid</i> rather than a decision: Theorem 7.2 and Corollary 7.3 settle the arithmetic of the sheep machine completely, leaving only its orbit question
G3	the paper’s main yield	Theorem 5.5 and Corollary 5.6 replace a bounded search (“no certificate at $m \leq 20,000$ ”) with an unconditional impossibility, <i>and</i> identify the mechanism (δ sweeps every modulus; β ’s parity governs the one 2-adic escape hatch); plus the barriers of Section 8
G4	demonstrated twice	the entire chain transfers to base 3 (Section 6) and to a second <i>wild</i> cryptid, the sheep machine (Section 7.1); Lemma 7.7 shows the saturation argument needs neither the family nor a base; the toolkit’s boundary is also mapped—two further machine families provably fall outside it

1.4 What this paper does not claim

We list these prominently because several are easy to over-read.

1. **No Turing machine is decided.** The 328 decided objects are arithmetic maps from a manufactured parameter box. Nothing here reduces the number of open busy-beaver holdouts.
2. **Neither flagship is resolved.** Whether the Space Needle’s reduction, or M3, halts remains open.
3. **Decided $\Rightarrow$ not hard.** A member admitting a separating congruence is, by that fact, outside the hard class. Our G2 contribution is a *partition* of a family, not progress on its hard members. Theorem 5.5 makes this sharper and less comfortable: the technique that decides 29.5% of the census is provably unavailable for the flagships, forever.

4. **The impossibility theorems are about one certificate class.** They say nothing about non-halting proofs of other kinds; the natural next class (automatic invariants) is discussed in Section 8 and is open.
5. **Completeness is relative to a box.** Theorem 4.4 is a statement about the enumerated parameter box, not about all odd- β members of the family.
6. **Universality of the family is open,** and Remark 8.2 shows one tempting route to settling it cannot work.

1.5 Epistemic conventions

Every numbered statement carries one of the following labels; the ledger in Section 9 lists all of them in one table.

[proved] A mathematical proof is given here or is routine and indicated. Where the proof contains a finite case analysis performed by computer, this is stated in the proof.

[proved, computer-assisted] The statement is equivalent to a finite computation, and a computer performed it exhaustively. This is a proof modulo correctness of the implementation; it is not a hand proof and we do not present it as one.

[machine-verified] Checked over an explicitly stated finite domain. *The general statement is not established.* Such a claim is evidence, not a theorem.

[measured] An empirical quantity (a count, a frequency, a growth rate).

[interpretation] A judgment, comparison or placement—for instance a claim that our equations lie outside a known decidable fragment. Not a formal result.

Each result is also accompanied by an **Evidence** note stating the exact domain over which computations were run. Threats to the validity of the computational component are discussed in Section 9.2 and are not negligible.

2 The family

Definition 2.1 (one-schema valuation machines). A machine is a tuple $(\alpha, \beta, \gamma, \delta, \varepsilon) \in \mathbb{Z}^5$. For $x \in \mathbb{N}$ write $x = 2^v u$ with u odd, $v = v_2(x)$. If $u = 1$ the machine **halts**. Otherwise $u = 2k + 1$ with $k \geq 1$, so $x = 2^{v+1}k + 2^v$, and

$$F(x) = A_v k + B_v, \quad A_v = \alpha 2^{v+1} + \beta, \quad B_v = \gamma 2^v + \delta v + \varepsilon.$$

The machine is *well defined* if F maps non-halting positive integers to positive integers.

The map reads the 2-adic valuation of its argument, chooses affine coefficients accordingly, and iterates; halting is landing exactly on a power of two. As a function of x the branch has slope $A_v/2^{v+1} \rightarrow \alpha$, so α is the asymptotic multiplier.

Proposition 2.2 (identification of the Space Needle’s reduction). *[PROVED] The member $(1, 3, 1, 1, 0)$ of Definition 2.1 coincides with Doucette’s reduction (1).*

Proof. With $b = 2^v(2k + 1)$ we have $m - 1 = 2k$, so $b + v + \frac{3}{2}(m - 1) = 2^{v+1}k + 2^v + v + 3k = (2^{v+1} + 3)k + (2^v + v)$, which is $A_vk + B_v$ for $(\alpha, \beta, \gamma, \delta, \varepsilon) = (1, 3, 1, 1, 0)$. The halting conditions agree by definition. $\square$

Evidence. Independently checked against an implementation of (1): zero mismatches for $2 \leq x < 300,000$; the orbit from $x_0 = 3$ is 3, 6, 10, 17, 41, 101, 251, $\dots$, which joins the published sequence at 6.

3 The census and exact congruence decision

3.1 The decision procedure

Fix a machine and a modulus $M \geq 2$. On branch v both the source $x = 2^{v+1}k + 2^v$ and the target $F(x) = A_vk + B_v$ are affine in k , so as k ranges over $\mathbb{Z}_M$ the pair traces the graph of one affine correspondence φ_v on $\mathbb{Z}_M$. Both A_v and $B_v \bmod M$ depend on v only through $(2^v \bmod M, v \bmod M)$, a state space of size at most M^2 ; so iterating v until that state repeats enumerates *every* branch, and $R_M = \bigcup_v \text{graph}(\varphi_v)$ is exactly computable. The orbit of x_0 lies in the R_M -closure of $x_0 \bmod M$, and the halting values have residues forming a computable eventually-periodic set H_M .

Definition 3.1. M *separates* a machine from x_0 if the R_M -closure of $x_0 \bmod M$ is disjoint from H_M . Such an M is a *congruence certificate* and proves non-halting from x_0 .

Proposition 3.2. [PROVED] *The procedure decides, for each M , whether M separates; hence it finds a separating modulus $\leq M_{\max}$ iff one exists. It is complete for its class: unlike a test on a finite orbit prefix, the closure is exact.*

Proof. Finiteness of the branch enumeration is the state-space argument above. The closure is taken over all k including $k = 0$, which only adds edges and hence only enlarges the closure, so a disjointness verdict remains sound. $\square$

3.2 Census results

Theorem 3.3 (census). [PROVED, COMPUTER-ASSISTED] *In the box $\alpha \in \{1, 2, 3\}$, $\beta \in \{-1, \dots, 7\}$, $\gamma \in \{1, 2, 3\}$, $\delta \in \{0, 1, 2\}$, $\varepsilon \in \{-2, \dots, 2\}$: of 1,080 tuples, 3 are not well defined and 1,077 are analysed. Exactly 318 (29.5%) admit a separating congruence at some $M \leq 64$ and hence provably never halt from $x_0 = 3$.*

Evidence. Every one of the 318 certificates was re-audited by an independent brute-force check over the integers ($x < 200,000$, powers of two enumerated completely, not reusing the branch enumeration that produced the certificate): **zero failures**. Smallest deciding moduli: 3 (41), 4 (105), 6 (68), 7 (3), 8 (18), 10 (34), 12 (17), 14 (2), 16 (14), 20 (6), 22 (1), 24 (2), 32 (1), 36 (1), 40 (2), 56 (3). [MEASURED]: decidability concentrates at even β (64–79% decided) against odd β (3.7–8.1%); Section 5 proves the mechanism behind this.

Theorem 3.4 (separation criteria at small moduli). [PROVED, COMPUTER-ASSISTED] *Separation at M depends on the parameters only through their residues mod M , so each criterion is a finite object on $(\mathbb{Z}_M)^5$. Write $A_0 = 2\alpha + \beta$, $B_0 = \gamma + \varepsilon$, $T = A_0 + B_0$.*

(T3) *Separation at 3 $\iff \delta \equiv 0, \gamma \equiv \alpha, \beta + \varepsilon \equiv 0 \pmod{3}$.*

(C4) *Separation at 4 $\iff \beta$ even and $2\alpha + \beta + \gamma + \varepsilon \equiv 3 \pmod{4}$.*

(C6) Separation at 6 $\iff$ one of: (a) A_0 even and $T \equiv 3 \pmod{6}$; (b) $A_0 \equiv 0, 4 \pmod{6}$ and $T \equiv 5 \pmod{6}$; (c) (T3) holds; (d) $A_0 \equiv B_0 \equiv 0 \pmod{6}$, $\delta \equiv 0 \pmod{3}$, $\gamma + \delta \equiv \beta + 5 \pmod{6}$.

(C8) At 8 only branches $v \in \{0, 1\}$ occur and A_0 must be even; separation is then determined by the forward orbit of 3 under an explicit four-node map.

Proof of (T3) and (C4), and status of (C6), (C8). Mod 3 the powers of two are $\{1, 2\}$, so a separating class is $\subseteq \{0\}$ and contains $x_0 = 3$, hence equals $\{0\}$; then $3 \mid x$ forces $k \equiv 1$ and $F(x) \equiv (-1)^v(\gamma - \alpha) + \beta + \delta v + \varepsilon$, whose vanishing at $v = 0, 1, 2$ gives the three congruences and conversely. Mod 4 the powers of two occupy $\{1, 2, 0\}$, so the only admissible class is $\{3\} \ni x_0$; only $v = 0$ has odd sources and $x \equiv 3 \pmod{4}$ iff k is odd, giving targets $A_0 + B_0$ and $3A_0 + B_0$, both $\equiv 3$ iff $2A_0 \equiv 0$ and $T \equiv 3$. For (C6) and (C8) we give the case decomposition in the repository; the statements as displayed are established by exhaustive verification on $(\mathbb{Z}_M)^5$, which is a complete finite check, and we therefore label them [PROVED, COMPUTER-ASSISTED] rather than [PROVED]. $\square$

Evidence. Exhaustive on $(\mathbb{Z}_M)^5$: 9/243 tuples separate at 3; 128/1,024 at 4; 1,308/7,776 at 6; 5,600/32,768 at 8; zero mismatches against the decision procedure in every case. Against the census: the criteria predict the decided sets exactly (41, 105, 68, 18 machines; zero discrepancies in either direction).

Remark 3.5. The Space Needle's reduction has $\delta = 1$ and so fails (T3) immediately. Section 5 upgrades this one-line observation into a theorem covering every modulus.

4 The last-step sieve, ten machines, and completeness

Since halting is landing on a power of two, one may ask which branches can be the *last* step. When branch v is expanding, its affine fixed point P_0/Q_0 gives the classical constraint: a halt can follow a v -step only if $Q_0 2^e \equiv P_0 \pmod{A_v}$ is solvable, where $P_0 = (2B_v - A_v)2^v$ and $Q_0 = 2^{v+1} - A_v$.

Lemma 4.1 (sieve lemma). [PROVED] *Let β be odd, so A_v is odd. For every member of the family and every expanding branch v ,*

$$\text{branch } v \text{ can immediately precede a halt} \iff B_v \in \langle 2 \rangle \pmod{A_v},$$

where $\langle 2 \rangle$ is the multiplicative group generated by 2.

Proof. Reduce mod A_v : the A_v -terms vanish, leaving $Q_0 \equiv 2^{v+1}$, a unit, and $P_0 \equiv 2B_v 2^v = B_v 2^{v+1}$. Hence $P_0 Q_0^{-1} \equiv B_v$, and solvability of $2^e \equiv P_0 Q_0^{-1}$ is exactly the stated membership. $\square$

Evidence. All 672 well-defined odd- β members of the box, every branch $v \leq 80$: 51,030 branches, zero mismatches against an independent implementation via the fixed point.

Corollary 4.2. [PROVED] *Since $\alpha 2^{v+1} \equiv -\beta \pmod{A_v}$, $\alpha \cdot 2B_v \equiv 2\alpha\delta v + 2\alpha\varepsilon - \beta\gamma \pmod{A_v}$: the sieve target is affine in v . The exponential appears only in the modulus, never in the target.*

Tractability of the membership question is governed by the size of $\langle 2 \rangle \pmod{A_v}$. Four (α, β) classes are *listable*— $\text{ord}_{A_v}(2)$ grows linearly in v , so the powers of two form a short explicit list—and the remainder are *thin*, with ord growing exponentially on average and no elementary handle. [MEASURED]

Theorem 4.3 (ten machines). [PROVED] *The following ten members never halt after their first step; equivalently their halting starts are exactly the powers of two, which halt at step 0. (Labels follow the program’s numbering, in which T3 is the criterion of Theorem 3.4 and not a machine.)*

<i>machine</i>	$2B_v \bmod A_v$	<i>why every branch is forbidden</i>	<i>label</i>
(1, 1, 2, 0, 1)	0	$B_v = A_v$; 0 is not a unit	T1
(2, -1, 2, 0, 1)	3	odd, $1 < 3 < A_v$ for $v \geq 1$	T9
(2, -1, 2, 1, 1)	$2v + 3$	odd, in range for $v \geq 1$	T2
(2, -1, 2, 2, 1)	$4v + 3$	odd, in range for $v \geq 2$	T4
(2, -1, 3, 0, 0)	$2^{v+1} + 1$	odd, in range for $v \geq 1$	T5
(2, -1, 3, 1, 0)	$2^{v+1} + 2v + 1$	odd, in range for $v \geq 2$	T6
(2, -1, 1, 2, -1)	$2^{v+1} + 4v - 2$	$= 2 \cdot \text{odd} < A_v$, so $2^v = 2 - 2v$: impossible	T10
(2, 1, 1, 2, -1)	$2^{v+1} + 4v - 2$	signed list; $4v - 2 = 2^{v+1}$: impossible	T7
(3, 3, 2, 0, 1)	$\equiv 0 \bmod d_v$	$B_v = d_v = 2^{v+1} + 1 \mid A_v$; powers of 2 are units mod d_v	T8
(3, 3, 3, 0, 0)	-3	$3 \mid A_v$, $2B_v = A_v - 3$; powers of 2 mod 3 are $\{1, 2\}$	T11

In each row the second column is an exact integer identity (e.g. $2B_v = A_v + 3$ for T9), the powers of two mod A_v form the explicit listable set of that machine’s class, and the stated parity, size or divisor argument excludes membership for all v beyond finitely many boundary branches, which land on 0.

Evidence. Closed forms asserted as integer identities to $v \leq 4,000$ on a second, independently written implementation; every branch $v \leq 200$ confirmed forbidden; brute force on all ten (every non-power-of-two start below 20,000, 400 steps): zero halts. Two of the ten upgrade earlier certificates from “never halts from $x_0 = 3$ ” to “from any start”; T10 has a genuine cycle ($F(3) = 3$) and still never halts from any non-power-of-two start.

Theorem 4.4 (completeness of the sieve method within the box). [PROVED, COMPUTER-ASSISTED] *Within the enumerated box, the ten machines of Theorem 4.3 are the only ones all of whose branches are forbidden. For every other well-defined odd- β member there is an explicit surviving-branch certificate: a branch $v \leq 22$ and an exponent e with $2^e \equiv 2B_v \pmod{A_v}$.*

Evidence. Of 672 machines: 10 all-forbidden, 620 with a surviving-branch certificate (each verified by modular exponentiation), 42 sieve-silent (the $(1, -1)$ class, which has no expanding branch).

A surviving branch is a positive witness, not a failure to find one; the method is therefore closed by a finite proof rather than by an exhausted budget. We stress two limits: the statement is relative to the box, and a surviving branch does not make a machine halt—it certifies only that this method cannot decide it.

For the Space Needle’s reduction the lemma specialises to a clean open condition. With $A_v = 2^{v+1} + 3$ we get $2^{v+1} \equiv -3$, so $\langle 2 \rangle = \{\pm 3^a 2^i\}$, and branch v can precede a halt iff

$$2v - 3 \in \{\pm 3^a 2^i\} \pmod{2^{v+1} + 3}. \quad (2)$$

Observation 4.5. [MEASURED] The set of branches satisfying (2) does not appear to thin out: 18 of the branches $v \leq 40$ survive, including $v = 34, 37, 38$. The weighted forbidden mass is 0.2868 and 19 of the first 35 valuations are forbidden, independently reproducing an earlier measurement. *We draw no asymptotic conclusion*; this is stated to record that no all-branches theorem is available for the Needle by this route.

5 No congruence certificate: the saturation theorems

We now show the certificate class of Section 3 is empty for the flagships—not below a bound, but outright. Write $M = 2^s M'$ with M' odd. For odd modulus the branch correspondence is a single affine map $\varphi_v(c) = \mu_v c + a_v$, and a short computation gives $\mu_v = \alpha + \beta 2^{-(v+1)}$ and $a_v = B_v - A_v/2 = (\gamma - \alpha)2^v + \delta v + (\varepsilon - \beta/2)$.

Theorem 5.1 (δ -saturation). *[PROVED] Let M be odd, $\gcd(\delta, M) = 1$ and $\gcd(\text{ord}_M(2), M) = 1$. Then for every residue c the one-step image $\{\varphi_v(c) : v \geq 0\}$ is all of $\mathbb{Z}_M$.*

Proof. On a class $v \equiv v_0 \pmod{\text{ord}_M(2)}$ the quantity 2^v is constant, so μ_v and $(\gamma - \alpha)2^v$ are constant and the only surviving v -dependence is δv : thus $a_v - a_{v_0} \equiv \delta(v - v_0)$. As v runs over the infinite progression $v_0 + \text{ord}_M(2)\mathbb{Z}$, the difference sweeps the subgroup generated by $\gcd(\text{ord}_M(2), M) = 1$, i.e. all of $\mathbb{Z}_M$, and $\gcd(\delta, M) = 1$ preserves this. $\square$

Since $\text{ord}_p(2) \mid p - 1$, the order hypothesis holds automatically at every odd prime; already this shows no odd prime separates the Needle's reduction. The hypothesis is scaffolding and is removed next.

Lemma 5.2 (descent). *[PROVED] Let M be odd with $\gcd(\delta, M) = 1$. Then the conclusion of Theorem 5.1 holds with no hypothesis on $\text{ord}_M(2)$.*

Proof. Set $g_0 = M$, $g_{j+1} = \gcd(\text{ord}_{g_j}(2), g_j)$, and let I_j be the image of $\{\varphi_v(c)\}$ in $\mathbb{Z}_{g_j}$. For odd $g \geq 3$, $\text{ord}_g(2) \leq \lambda(g) < g$, so the chain strictly decreases to 1. Mod g_j with $h = \text{ord}_{g_j}(2)$: on each class $v \equiv w \pmod{h}$ everything but δv is constant, and δv sweeps the *full* subgroup $\langle g_{j+1} \rangle$ —here the infinitude of the progression is essential, one period being insufficient. So I_j is a union of h cosets of $\langle g_{j+1} \rangle$, and reducing mod g_{j+1} annihilates the sweep and leaves exactly the base points, whence $I_j = \mathbb{Z}_{g_j} \iff I_{j+1} = \mathbb{Z}_{g_{j+1}}$. The chain bottoms out at $g = 1$, where fullness is trivial; walking back up gives $I_0 = \mathbb{Z}_M$. $\square$

Evidence. 2,500 (machine, odd M , c) instances with $\gcd(\delta, M) = 1$, of which 708 have $\gcd(\text{ord}_M(2), M) > 1$ —exactly the cases the scaffold hypothesis excluded—the one-step image fell short of $\mathbb{Z}_M$ in none. The chain strictly decreases for every odd $g < 6,000$, longest chain length 7.

5.1 Even moduli: the parity escape hatch

A residue $c \bmod 2^s$ with $v_2(c) = v < s$ *pins the branch index*: the low bits of the value are the valuation. That is precisely the information an odd modulus cannot carry, and it is the entirety of what an even modulus buys. On a source $\equiv 0 \pmod{2^s}$ every branch $v \geq s$ is available, k is free mod 2^s , and since $v + 1 > s$,

$$F(x) \equiv \beta k + \delta v + \varepsilon \pmod{2^s}.$$

If β is odd, βk sweeps all of $\mathbb{Z}_{2^s}$ and the 2-adic information is destroyed in one step; if β is even only $\langle \gcd(\beta, 2^s) \rangle$ is reached—the escape hatch.

Theorem 5.3 (even-modulus saturation). *[PROVED] Let $M = 2^s M'$, β odd, $\gcd(\delta, M') = 1$, and suppose the R_M -closure of x_0 contains some $c_0 \equiv 0 \pmod{2^s}$. Then that closure is all of $\mathbb{Z}_M$.*

Proof. From c_0 every branch $v \geq s$ is available (choose an odd representative mod M' , possible since M' is odd and 2^v is invertible there). Splitting the source congruence by CRT: mod 2^s it holds identically, so k is unconstrained; mod M' it determines k . The target's 2-part sweeps $\mathbb{Z}_{2^s}$ by the display above; its M' -part, over $v \geq s$, is the δ -sweep of Lemma 5.2 restricted to $v \geq s$, which costs nothing since each order class contains infinitely many such v . The product is $\mathbb{Z}_{2^s} \times \mathbb{Z}_{M'} = \mathbb{Z}_M$. $\square$

Evidence. Proof steps checked on 3,000 instances; the conclusion on 7,840 (odd- β machine, even modulus) pairs: zero failures. *Falsifier:* each of the census's 274 even-modulus certificates must violate a hypothesis—267 have β even (they spend exactly the hatch) and 7 have $\delta = 0$; none would refute. The 44 odd-modulus certificates all violate $\gcd(\delta, M) = 1$.

5.2 The reach lemma

It remains to supply the c_0 of Theorem 5.3.

Lemma 5.4 (reach). *[PROVED] Let β be odd and $A_v > 0$ for all v . For every residue c mod $M = 2^s M'$ there is a lift $x \equiv c \pmod{M}$ whose F -orbit reaches a value $\equiv 0 \pmod{2^s}$ within $s + 1$ steps.*

Proof (exponent ledger). If $c \equiv 0$ there is nothing to prove; else let $v_0 = v_2(c \bmod 2^s) < s$ and take $x = c + 2^s M' T$ with T free. Iterating F *exactly* keeps the T -dependence affine, $y_n = p_n + q_n T$, with $q_n = A_{v_{n-1}} \cdots A_{v_0} M' 2^{e_n}$ and the ledger $e_1 = s - (v_0 + 1)$, $e_{n+1} = e_n - (v_n + 1)$: forming $k = (y_n - 2^{v_n}) / 2^{v_n+1}$ divides the T -coefficient by 2^{v_n+1} , and multiplying by the odd A_{v_n} changes nothing 2-adically. Two events are exhaustive. *Done-event:* some p_n is divisible by 2^{e_n} , and then $y_n \equiv 0 \pmod{2^s}$ has a solution $T \equiv -(p_n / 2^{e_n})(q_n / 2^{e_n})^{-1} \pmod{2^{s-e_n}}$, which may be taken arbitrarily large. *Ledger-event:* e reaches 0, so k is a unit times $T \bmod 2^s$ and, A_v being odd, $y_{n+1} = A_v k + B_v$ takes every residue mod 2^s including 0. While neither has fired, $v_2(p_n) < e_n$ makes v_n independent of T , so the branch sequence is well defined, and the ledger strictly decreases from at most $s - 1$; the process ends within $s + 1$ steps. For T large every y_n is positive with odd part exceeding 1, so no step halts en route. $\square$

Evidence. Constructive: the recipe was used to *build* the witness lift and its orbit was *run*, on 4,000 (machine, modulus, residue) instances—a hit on $0 \bmod 2^s$ every time, deepest at 9 steps against the bound 10. The ledger bookkeeping was checked in isolation on 3,000 chains and the T -independence of the branch sequence confirmed. *Falsifier:* β -even machines with even-modulus certificates are 2-adically confined, so reach must fail for them—it does, in 40 of 40 tested.

Theorem 5.5 (no congruence certificate). *[PROVED] Let a member have β odd, $A_v > 0$, and $\gcd(\delta, M') = 1$ where M' is the odd part of M . Then the R_M -closure of every residue is all of $\mathbb{Z}_M$: no modulus M separates, and no branch can be excluded.*

Proof. Odd M is Lemma 5.2. For even M , Lemma 5.4 places a residue $\equiv 0 \pmod{2^s}$ in the closure and Theorem 5.3, with Lemma 5.2 supplying its M' -part, finishes. $\square$

Corollary 5.6. *[PROVED] No congruence certificate at any modulus separates the Space Needle's reduction. ($\beta = 3$ odd, $\delta = 1$, $A_v = 2^{v+1} + 3 > 0$, so every hypothesis holds at every M .) Equivalently: the technique that decides 29.5% of the census can never decide this instance.*

Evidence. 7,616 (odd- β machine, modulus) pairs meeting the hypotheses: closure short of $\mathbb{Z}_M$ in none. The Needle’s reduction directly at every modulus $2 \leq M \leq 400$: saturated in all cases.

What began as “ δ blocks the cheapest certificate” (the one-line failure of (T3)) is now: δ blocks *every* certificate at *every* modulus, and the escape hatch that accounts for 267 of the 274 even-modulus certificates is closed exactly by β odd.

6 Transfer to base q : machine M3

None of the proofs above uses base 2 essentially. We demonstrate the transfer on a second machine. **M3** is the six-rule system

$$\begin{aligned} A(1, 3k) &\rightarrow \text{HALT}, & A(1, 3k+1) &\rightarrow A(3k+4, 1), & A(1, 3k+2) &\rightarrow A(3k+3, 2), \\ A(3k, b) &\rightarrow A(k, b+2k+1), & A(3k+1, b) &\rightarrow A(4k+b+3, 1), & A(3k+2, b) &\rightarrow A(4k+b+5, 1), \end{aligned}$$

started at $A(1, 1)$; its halting is equivalent to the orbit of the return map to the $b = 1$ section reaching a power of 27, which is open.

Proposition 6.1 (M3’s branch table). *[MACHINE-VERIFIED] Writing a non-spine value as $a = 3^{j+1}m + r3^j$ with $j = v_3(a)$ and leading digit $r \in \{1, 2\}$ (the spine being the pure powers 3^j , which halt iff $3 \mid j$), the return map is*

$$G(a) = A_j m + B_{j,r}, \quad A_j = 3^{j+1} + 1, \quad B_{j,r} = r3^j + j + c_r, \quad c_1 = 3, \quad c_2 = 4,$$

reached in $j+1$ raw steps. In the coordinates of Definition 2.1 read base 3: $\alpha = 1$, $\beta = 1$, $\gamma = r$, $\delta = 1$, $\varepsilon_r = r + 2$.

Evidence. Checked against the raw six rules on 59,988 non-spine values $a < 60,000$: zero mismatches, including the step count $j+1$; spine behaviour exact for $j = 1, \dots, 11$. We label this *[MACHINE-VERIFIED] deliberately*: the induction over the j divide-steps is routine but is not written out here, and Theorems 6.3 and 6.4 below inherit that dependency.

Lemma 6.2 (base- q sieve lemma). *[PROVED] For a branch sending $x = q^w m + s$ to $Am + B$ with $\gcd(q, A) = 1$ and halting set $H = \{cp^t\}$, $\gcd(\rho, A) = 1$: the branch can immediately precede a halt iff $B \in c\langle\rho\rangle \pmod{A}$. For M3 the fixed point gives $Q_0 = 3^{j+1} - A_j = -1$ exactly, so the condition reads $27^t \equiv B_{j,r} \pmod{A_j}$.*

The base-2 form of Lemma 4.1 (with its factor 2) was a normalisation artifact; the invariant content in every base is *target* $\equiv B$.

Theorem 6.3 (M3: last two steps pinned). *[PROVED, COMPUTER-ASSISTED] For every branch index j : a halt of M3 can only follow a branch with $v_3 = 1$, and the same holds one step earlier. The excluded-pair frequency $1 - (2/9)^2 = 77/81$ is exact. Both surviving depth-one branches are realised: $21 \rightarrow 27$ and $478293 \rightarrow 27^4$.*

Evidence. Depth-one verdicts agree with an independent solver on 50 branches, survivors exactly (1, 1) and (1, 2). The depth-two argument (a parity lemma removing even j ; for odd $j \geq 5$ the targets lie strictly inside the gaps of $\{\pm 3^i\}$; small j checked exactly) was machine-checked to $j = 2,000$ and cross-checked against an exact finite test for $2 \leq j \leq 500$. *Caveat*: the present author’s independent re-check of this theorem covered only its brute-force consequence (among all non-spine $a < 10^6$, exactly three starts halt within two return steps—15, 21, 478293—with every branch value en route having $v_3 = 1$), not the depth-two closed forms. This is the least independently corroborated result in the paper.

Theorem 6.4 (M3: no congruence certificate). [PROVED] (modulo Proposition 6.1) For every modulus $M \geq 2$ the closure of every residue under M3’s branch relation is all of $\mathbb{Z}_M$; no congruence certificate at any modulus separates M3’s orbit from the residues of $\{27^k\}$.

Proof. The chain of Section 5 with $2 \rightarrow 3$: “odd M ” becomes “ M coprime to 3”; “ β odd” becomes “ β coprime to 3” (here $\beta = 1$); the exponent ledger spends trits. M3 has $\delta = 1$, so all hypotheses hold at every modulus. Spine transitions only enlarge closures. $\square$

Evidence. Independent implementation: every modulus $2 \leq M \leq 250$, three start residues each (747 closures), all equal to $\mathbb{Z}_M$.

Remark 6.5 (the flagship dichotomy). [PROVED] Both flagships satisfy Theorem 5.5’s hypotheses for the same reason ($\delta = 1$, β coprime to the base) and so resist every congruence certificate. What separates their fates under the *sieve* is one class parameter, the rank of $\langle q \rangle \bmod A_v$: M3’s class forces $q^{j+1} \equiv -1$, making $\langle 3 \rangle = \{\pm 3^i\}$ listable (rank one), so the gap argument closes every branch question; the Needle’s class gives $\langle 2 \rangle = \{\pm 3^a 2^i\}$ (rank two, thin), which is the open condition (2). In the census, 120 members share the Needle’s group.

Remark 6.6 (interface boundary). [INTERPRETATION] The toolkit requires a valuation-indexed branch family together with a halting set defined by a condition on the *value*. Two further machine families studied in the program—the Hydra/Antihydra maps and the FRACTRAN cryptid Fenrir—fall outside it: their branch modulus is constant and halting is a condition on a path counter rather than on the value, so the sieve has no object to apply to. We record this as a mapped boundary, not a failure.

7 Two tests outside the design set

Everything above was developed against the Space Needle’s reduction. That is a real threat to validity, and this section reports the two tests we can run against it: a second *wild* cryptid that lies inside the interface, and an old case-file machine that lies outside the family but inside the saturation argument.

7.1 The sheep machine: a second wild instance

The Turing machine 1RB1LA_OLCORC_1LE1RD_1RE1RC_1LFOLA_---1LE was found by *sheep* on 7 April 2026 and is listed as a Cryptid on the bbchallenge wiki, which states the halting-equivalent reduction

$$f(n) = \begin{cases} \text{HALT} & \text{oddPart}(n) = 1, \\ n + v_2(n) + 3 & \text{oddPart}(n) = 3, \\ n + v_2(n) + \frac{\text{oddPart}(n)-1}{2} & \text{oddPart}(n) > 3, \end{cases} \quad \text{started at } n = 5,$$

together with two lemmas: that $m2^a + a + \frac{m-1}{2}$ is never a power of two for $a \geq 2$, $m \geq 5$; and that $3 \cdot 2^a + a + 3$ is never a power of two.

Proposition 7.1 (identification). [PROVED] On the generic branch ($\text{oddPart}(n) > 3$), writing $n = 2^v m$ with $m = 2k + 1$,

$$f(n) = A_v k + B_v, \quad A_v = 2^{v+1} + 1, \quad B_v = 2^v + v,$$

so the generic branch is the census member $(\alpha, \beta, \gamma, \delta, \varepsilon) = (1, 1, 1, 1, 0)$. The Needle’s reduction is $(1, 3, 1, 1, 0)$: the two machines differ in β alone.

Evidence. 15,920 systematic and 200,000 random (v, k) pairs with $v < 200$, $k < 10^{30}$: no mismatch. For $n < 60,000$, f differs from the census member exactly on the values with $\text{oddPart}(n) = 3$.

The exceptional branch is not decoration. The census records member $(1, 1, 1, 1, 0)$ as *halting*: $3 \mapsto 4 = 2^2$. The sheep machine's third branch intercepts precisely that value, sending $3 \mapsto 6$. One extra branch on one odd part converts a halting census member into an open problem—the sharpest statement we can make about what a manufactured box misses.

Theorem 7.2 (the sieve closes). [PROVED] *For the sheep machine, branch v can immediately precede a halt if and only if $v \in \{0, 1\}$.*

Proof. A generic step lands on a power of two iff $2^t \equiv B_v \pmod{A_v}$ for some t , i.e. iff $B_v \in \langle 2 \rangle$. Since $A_v = 2^{v+1} + 1$ we have $2^{v+1} \equiv -1$ and $2^{-1} \equiv 2^v + 1$, so $B_v = 2^{-1} - 1 + v$ and the condition is

$$2^{t+1} \equiv 2v - 1 \pmod{2^{v+1} + 1}.$$

Also $2^{v+1} \equiv -1$ forces $\langle 2 \rangle \subseteq \{2^i\}_{i \leq v} \cup \{A_v - 2^i\}_{i \leq v}$. If $2v - 1 = 2^i$ then $2v - 1$ odd gives $i = 0$, hence $v = 1$. If $2v - 1 = A_v - 2^i$ then $2^i = 2^{v+1} - 2v + 2$, which for $v \geq 3$ lies strictly between 2^v and 2^{v+1} (as $2^v > 2v - 2$ and $2v > 2$) and so is not a power of two; $v = 2$ gives 6. Finally $v = 0$: $A_0 = 3$ and $2v - 1 \equiv 2 = 2^1 \in \langle 2 \rangle$. $\square$

The wiki's first lemma is the $v \geq 2$ half of Theorem 7.2 with the threshold $a \geq 2$ assumed; here the threshold is a conclusion, and the two surviving branches are located at the same time. The second lemma is recovered the same way: for $a \geq 3$, $3 \cdot 2^a + a + 3$ lies strictly between $3 \cdot 2^a$ and 2^{a+2} , and $a \leq 2$ gives 6, 10, 17.

Corollary 7.3 (the halting set in closed form). [PROVED] *A value halts in one step iff it is a power of two, and reaches a power of two in exactly one step iff it lies in*

$$H_0 = \left\{ \frac{2^{2j+1}+1}{3} \right\}_{j \geq 2} = 11, 43, 171, 683, \dots \quad \text{or} \quad H_1 = \left\{ \frac{2(2^{4j}-1)}{5} \right\}_{j \geq 2} = 102, 1638, 26214, \dots$$

so $H = \{2^i\} \cup H_0 \cup H_1$.

Evidence. Brute force over $n < 300,000$ returns exactly the 11 predicted members. The excluded first member of each family is $m = 3$ in both cases—the exceptional branch again.

Finally $\beta = 1$ is odd and $\delta = 1$, so Theorem 5.5 applies verbatim: no congruence certificate at any modulus separates the sheep machine. *Evidence.* Direct closure computation gives all of $\mathbb{Z}_M$ for every $M \leq 200$.

Remark 7.4 (the dichotomy, tested). [INTERPRETATION] Remark 6.5 predicts that the last-step sieve closes when $\langle 2 \rangle$ modulo A_v is a rank-one listable group and stalls when it is rank two and thin. The sheep machine has $A_v = 2^{v+1} + 1$, hence $\langle 2 \rangle = \{\pm 2^i\}$, rank one; the Needle has $A_v = 2^{v+1} + 3$, hence $\{\pm 3^a 2^i\}$, rank two. The sieve closes for the former and not the latter. This is the first test of that prediction on a machine we did not construct, and it was on record before the sheep machine was examined.

The result is a cryptid whose *arithmetic* is settled completely—we know exactly which values can halt, in closed form—and whose orbit question is untouched. The Needle is one where neither is settled. We regard this as the paper's clearest evidence that the sieve-group rank, and not the machine, is what governs the last-step analysis.

Depth two, and why depth does not help

The same reduction runs one step further. Since $3N = 2^{2j+1} + 1$ and $5N = 2^{4j+1} - 2$ identically, “ $N \equiv B_v \pmod{A_v}$ ” clears denominators into $2^{2j+1} \equiv 3B_v - 1 \pmod{3A_v}$ and $2^{4j+1} \equiv 5B_v + 2 \pmod{5A_v}$; the larger modulus is required because $3 \mid A_v$ whenever v is even.

Theorem 7.5 (depth two). *[PROVED] No branch $v \geq 6$ can be the second-to-last step before a halt; the depth-two survivor set is exactly $\{0, 1, 2, 3, 5\}$.*

Proof. Reduce each condition modulo A_v (a necessary condition). Because $2^{v+1} \equiv -1$, $\text{ord}_{A_v}(2)$ is even, so the exponent parity of each element of $\langle 2 \rangle = \{2^i\}_{i \leq v} \cup \{A_v - 2^i\}_{i \leq v}$ is well defined, and the two targets reduce to $2^v + 3v - 2$ and $2^v + 5v$. Each must be $\pm 2^i$. For H_0 : $2^v < 2^v + 3v - 2 < 2^{v+1}$ kills the first case for $v \geq 4$, and the second gives $2^i = 2^v - 3v + 3 \in (2^{v-1}, 2^v)$, impossible for $v \geq 5$. For H_1 : $2^v < 2^v + 5v < 2^{v+1}$ for $v \geq 5$, and $2^i = 2^v - 5v + 1$ lies in $(2^{v-1}, 2^v)$ for $v \geq 6$ —while $v = 5$ gives $32 - 25 + 1 = 8 = 2^3$, a genuine solution. Branches $v \leq 5$ are then checked directly. $\square$

Iterating is possible in closed form. Call $N(i) = (2^{\alpha+ei} + b)/c$ with c odd a *geometric family*; the halting set is one such family, and the shape is closed under preimages, since $f(x) = N(i)$ for $x = 2^v(2k+1)$ reduces to $2^{\alpha+ei} \equiv cB_v - b \pmod{cA_v}$, whose solutions form a class $i \equiv i_0 \pmod{P}$, and substituting back yields a family with $e' = eP$, $c' = cA_v$. So each depth is computed exactly, with no search over x .

depth	survivor set	families	forbidden branch mass	admissible word mass
1	$\{0, 1\}$	2	0.250000	0.750000
2	$\{0, 1, 2, 3, 5\}$	7	0.046875	0.714844
3	$\{0, \dots, 6\}$	23	0.007812	0.709259
4	$\{0, \dots, 6, 9\}$	90	0.006836	0.704411
5	$\{0, \dots, 10, 13\}$	346	0.000427	0.704110
6	$\{0, \dots, 13\}$	1421	0.000061	0.704067

Remark 7.6 (the sieve saturates). *[MEASURED]* The forbidden mass per depth collapses geometrically while the family count grows by a factor ≈ 3.8 per depth, so the admissible word mass is a *convergent* product, measured at 0.704067 by depth six and extrapolating to ≈ 0.70406 . The last-step sieve, at any depth, can therefore never forbid more than about 29.6% of branch words. We note that the depth-five data alone extrapolated to 0.704045, and depth six then measured 0.704067; the extrapolation was accurate to five decimals before the check existed. Depth seven was not attempted: cost grows by a factor ≈ 15 per depth (1 s, 80 s, 6.4 h for depths four to six).

The mechanism is legible: the depth- d target is a union of F_d families, so branch v survives if *any* is reachable; each is reachable with probability $\approx \text{ord}(2)/A_v = O(v)/2^v$, so v survives up to $v \approx \log_2 F_d$, which grows linearly in d . Survivor sets grow, the forbidden mass decays geometrically, and the product converges. *Depth is not a resource that buys arbitrarily much*—which, with Theorem 5.5, closes both congruence-flavoured lanes on this machine with numbers attached, and leaves automatic invariants as the only one untried. Evidence: ground truth by brute force to 3×10^6 agrees at depths one to four; the recursion is verified against f itself on two members of every family generated.

7.2 A machine-independent saturation lemma

The descent of Lemma 5.2 does not need the family at all.

Lemma 7.7 (linear-exponential saturation). *[PROVED] Let $M \geq 1$ and let $A, \delta, \varepsilon, \kappa, \rho$ be integers with $\gcd(\delta, M) = \gcd(\rho, M) = 1$. Put $f(n) = \delta n + \varepsilon + \kappa \rho^n$ and $R = \{(c, Ac + f(n)) \bmod M : n \geq n_0\}$. Then from every $c \in \mathbb{Z}_M$, the residues reachable in at least one R -step are all of $\mathbb{Z}_M$. There is no hypothesis on A , none on κ , and n_0 is irrelevant.*

Proof. Strong induction on M ; $M = 1$ is trivial. Let $T = \text{ord}_M(\rho)$ and $g = \gcd(T, M)$; as $T \leq \varphi(M) < M$ we have $g < M$. For any c and n_1 , the successors $\{Ac + f(n_1 + jT)\}_{j \geq 0} = Ac + f(n_1) + \delta T\mathbb{Z} = Ac + f(n_1) + \langle g \rangle$, using $\gcd(\delta, M) = 1$; so every residue reached in ≥ 1 step lies in a full coset of $\langle g \rangle$ that is also reached. If $g = 1$ we are done. Otherwise $\pi : \mathbb{Z}_M \rightarrow \mathbb{Z}_M / \langle g \rangle \cong \mathbb{Z}_g$ carries R to a relation of the same shape with $\gcd(\delta, g) = \gcd(\rho, g) = 1$; by induction π of the reachable set is all of $\mathbb{Z}_g$, so the reachable set meets every coset and hence, by the previous sentence, contains each of them. $\square$

Evidence. 1,705 random parameter tuples with $M < 400$ all saturate; dropping $\gcd(\delta, M) = 1$ makes 437 of 671 instances fail, so the hypothesis is load-bearing.

Applying it to a machine outside the family: the counter system **M1** studied earlier in this project reduces to a return map F whose dominant branch word satisfies, for $n \geq 1$ and exactly the D with $16 \cdot 2^n - 2n - 10 \leq D \leq 20 \cdot 2^n - 2n - 13$,

$$F(D) = 16D - 240 \cdot 2^n + 32n + 169 \quad ([\text{PROVED}])$$

(both interval endpoints derived: the lower from the exit guard, the upper from n being the cascade length). That interval has length $4 \cdot 2^n - 2$, so once $2^n \geq (M+2)/4$ it contains a complete residue system modulo M , and therefore the true edge relation of F modulo M contains the relation of Lemma 7.7 with $A = 16$, $\delta = 32$, $\varepsilon = 169$, $\kappa = -240$, $\rho = 2$.

Theorem 7.8 (M1). *[PROVED] No congruence certificate at any odd modulus proves M1 non-halting; this already follows from its single dominant branch word.*

Evidence. Verified exhaustively for every odd $M \leq 401$ and for 25 random odd $M < 1200$. On the even side the orbit is confined to one class modulo 16 and no 2^e with $e \geq 5$ says more, so the complete congruence content of M1 is a single residue class—which does not separate it from its halting set. M1 is thus the first machine we place on the β -even side of the parity dichotomy of Section 5.1.

8 Expressiveness barriers

Whether the family is Turing-complete is open. This section bounds what such a proof must overcome and refutes one tempting genre of impossibility argument.

Theorem 8.1 (branch rigidity). *[PROVED] Branch v sends every non-halting input to the same valuation iff $v_2(B_v) < v_2(A_v)$, and then $v_2(F(x)) = v_2(B_v)$ throughout. If this holds at every v , the branch sequence of every orbit is a fixed eventually-periodic sequence independent of the state; the orbit is a periodic composition of affine maps with closed form $x_n = c\lambda^n + d$, and the machine performs no state-dependent branching.*

Evidence. 6,462 (machine, branch) pairs, zero disagreements. [MEASURED]: in the box $\alpha \in [0, 3]$, $\beta \in [-4, 8]$, $\gamma, \delta, \varepsilon \in [-3, 4]$ (19,092 well-defined members), exactly 2,351 (12.31%) are globally rigid, all with β even. A closed-form criterion is exact for β even and nonzero; the case $\beta = 0$ is genuinely different ($v_2(A_v) = v + 1$ unbounded makes rigidity generic) and supplies 758 of the 2,351.

Remark 8.2 (the counting genre cannot work). $(A_v, B_v) \bmod M$ is eventually periodic in v [PROVED], and the achievable branch tables form a rank-5 subgroup among tables of length V [PROVED]. It is tempting to conclude the family is too poor to host a Conway-style reduction, which needs M independent branches at modulus M . That conclusion is unsupported: measured branch *bandwidth*—distinct $(A_v, B_v) \bmod M$ actually achieved—is at least M for every modulus we tested (100 at $M = 23$, 412 at $M = 101$, 1,036 at $M = 257$) [MEASURED]. Moreover the genre cannot be repaired [INTERPRETATION]: universality is a property of a *single point*, not of a family—a five-parameter family may contain a universal member because a universal machine is a zero-parameter object—so no dimension, rank, density or bandwidth argument can exclude it. What survives is a compression bound [PROVED]: at most $\lfloor \log_2(|\beta|/\rho) \rfloor$ branches are pairwise ρ -separated in slope, so k separated branches cost $|\beta| \geq \rho 2^{k-1}$.

Remark 8.3 (automatic invariants). [PROVED, COMPUTER-ASSISTED] for each stated size. Bounded-size searches for 2-automatic invariants separating the orbit from the halting set return none for the Needle’s reduction at ≤ 11 states (least-significant-bit-first, minimal-word), ≤ 13 (LSB, zero-invariant) and ≤ 13 (most-significant-bit-first), and none for M3 in base 3 at ≤ 7 states; encodings were shown non-vacuous by planting certificates at matched sizes. Unbounded size is open, and after Theorem 5.5 this is the sharpest open certificate question for the family. Known non-definability results [DP26] constrain the reachability *relation* of $3x+1$ -type maps in Büchi arithmetic but do not obstruct invariants.

9 Status of every result, and threats to validity

9.1 The ledger

result	status	note
Prop. 2.2 identification	[PROVED]	one line of algebra
Prop. 3.2 exactness	[PROVED]	
Thm 3.3 census	[PROVED, COMPUTER-ASSISTED]	each of 318 decisions is an exact finite closure computation; all re-audited
Thm 3.4 (T3), (C4)	[PROVED]	hand proofs given
Thm 3.4 (C6), (C8)	[PROVED, COMPUTER-ASSISTED]	exhaustive on $(\mathbb{Z}_M)^5$; hand case analysis in repository only
Lem. 4.1 sieve lemma	[PROVED]	three-line reduction
Cor. 4.2 linear form	[PROVED]	
Thm 4.3 ten machines	[PROVED]	closed forms are exact identities; finitely many boundary branches checked
Thm 4.4 completeness	[PROVED, COMPUTER-ASSISTED]	finite: ten proofs plus 620 witnesses; relative to the box
Obs. 4.5 no thinning	[MEASURED]	no asymptotic claim made
Thm 5.1 δ -saturation	[PROVED]	
Lem. 5.2 descent	[PROVED]	
Thm 5.3 even saturation	[PROVED]	
Lem. 5.4 reach	[PROVED]	construction also executed
Thm 5.5, Cor. 5.6	[PROVED]	the paper's main result
Prop. 6.1 M3 branch table	[MACHINE-VERIFIED]	induction not written out; Thms 6.3, 6.4 depend on it
Lem. 6.2 base- q lemma	[PROVED]	
Thm 6.3 M3 two-step pinning	[PROVED, COMPUTER-ASSISTED]	least independently corroborated result here
Thm 6.4 M3 no-certificate	[PROVED]*	modulo Prop. 6.1
Rem. 6.5 dichotomy	[PROVED]	
Rem. 6.6 boundary	[INTERPRETATION]	
Prop. 7.1 sheep identification	[PROVED]	one line of algebra; broadly machine-checked
Thm 7.2 sheep sieve closes	[PROVED]	recovers and sharpens a community lemma
Cor. 7.3 sheep halting set	[PROVED]	matched to brute force below 3×10^5
Thm 7.5 sheep depth two	[PROVED]	survivor set exactly $\{0, 1, 2, 3, 5\}$
Rem. 7.6 sieve saturates	[MEASURED]	five depths; the convergence is extrapolated, not proved
Rem. 7.4 dichotomy tested	[INTERPRETATION]	the prediction was on record first; one instance
Lem. 7.7 linear-exponential saturation	[PROVED]	no hypothesis on the multiplier
Thm 7.8 M1 no odd certificate	[PROVED]	the even-side statement accompanying it is [MACHINE-VERIFIED]
Thm 8.1 rigidity	[PROVED]	the 12.31% figure is [MEASURED]
Rem. 8.2 counting genre	mixed	periodicity [PROVED]; bandwidth [MEASURED]; the single-point argument [INTERPRETATION]
Rem. 8.3 automatic invariants	[PROVED, COMPUTER-ASSISTED]	per stated size only
§11 placement	[INTERPRETATION]	a syntactic comparison, not a theorem

9.2 Threats to validity

1. **All verification is by our own implementations.** Several headline results were checked on two independently written code paths, and where possible against pre-existing implementations written for other purposes; but there is no third-party or formally verified check. A systematic misunderstanding of the semantics would not be caught. A Lean formalisation of Lemma 4.1, Lemma 5.2 and Lemma 5.4 is the natural remedy and is not yet done.
2. **[proved, computer-assisted] results depend on implementation correctness**, including correctness of the SAT encodings in Remark 8.3. Encoding adequacy was established by planting certificates, which addresses vacuity but not all forms of error.
3. **Proposition 6.1 is machine-verified, not proved**, and two theorems depend on it.
4. **Theorem 6.3’s depth-two argument received the weakest independent scrutiny** of any result claimed here; see its Evidence note.
5. **Selection effects in the census.** The box was chosen for computational convenience. Frequencies such as “29.5% decided” are properties of that box, not of the family.
6. **Developed against one machine—now partly mitigated.** Every theorem in Sections 4–6 was developed with the Space Needle in view, and a theory fitted to one instance can encode that instance’s accidents. Section 7 is the corrective: one wild cryptid found independently (sheep) falls inside the interface and confirms a prediction made beforehand, and one machine outside the family (M1) is nonetheless covered by the saturation argument. Two instances is better than one; it is not a demonstration of generality.
7. **Asymmetry of search evidence.** Refuting a certificate at a given size is cheap; finding one is expensive. Our machinery therefore produces negative results far more readily than positive ones—a property of the evidence, not of the machines.

10 Progress against the goals, revisited

G1 (observe): substantial. Lemma 4.1 and Corollary 4.2 give the family’s last-step sieve in closed form and isolate why it is hard—the exponential is in the modulus, not the target. Theorem 4.4 is, to our knowledge, unusual in kind: a method closed by a completeness statement rather than an exhausted budget. Theorem 3.4 supplies closed-form separation criteria where only $m = 3$ was previously characterised, and Remark 6.5 explains a difference between two machines that had previously only been observed.

G2 (decide): moved, and not on the target class. We decide $318 + 10$ members. Every one of them is a Collatz-like function from a manufactured box; none is a Turing machine; and each is decided precisely because it admits a certificate, which is what it means not to be hard. The flagships remain open. We regard the honest statement of this contribution as: *the census partitions a family into an easy part, a characterised hard part, and—via Theorem 5.5—a proof that the boundary between them is not an artifact of effort*. It should also be said plainly that Theorem 5.5 makes G2 harder, not easier: it removes the cheapest technique from consideration for the hard members permanently.

G3 (explain): the paper’s main yield. Before this work the statement was “no separating congruence was found for the Needle’s reduction at $m \leq 20,000$ ”—a bounded search, silent beyond its budget. Corollary 5.6 replaces it with an unconditional impossibility, and Sections 5–8 supply mechanisms rather than only outcomes: δ sweeps every modulus; β ’s parity governs the sole 2-adic escape hatch; rigidity marks off a slice that cannot compute; and one genre of universality barrier is refuted with a reason that applies to all its members.

G4 (transfer): demonstrated three times, with its boundary. Section 6 moves the whole chain to base 3 and obtains Theorem 6.4 for a second machine. Section 7.1 applies it unchanged to a cryptid found by someone else, and Section 7.2 strips the saturation argument of the family schema entirely, then applies it to a machine of a different kind (Theorem 7.8). Equally informative is Remark 6.6: two further families provably lie outside the toolkit’s interface. Three transfers is evidence of generality, not proof of it.

What did not move. No progress was made on either flagship’s actual halting question; the sheep machine’s orbit question is equally untouched, and the fact that its *arithmetic* closes completely while its orbit question does not is itself a reminder of how little the last-step analysis reaches; no positive non-halting theorem was obtained for any hard instance by any method; and the universality question for the family remains open, with one route now closed.

11 Discussion: where the difficulty now sits

Combining Lemma 4.1 with Theorem 5.5: every congruence-flavoured route to the Needle’s reduction is closed, and the halting condition itself is the two-generator membership (2). Lifting to the integers, a halt out of branch v at value 2^E requires $A_v k = 2^E - B_v$ with $k \geq 1$: a linear-exponential Diophantine condition in which the unknown multiplier k appears *linearly* against the exponentials 2^{v+1} and 2^E .

Observation 11.1. [INTERPRETATION] Systems of linear-exponential equations over $\{2, 3\}$ without such mixed products are decidable [DS25, CMS24]. The system above differs from those fragments in two syntactic respects: one product of a free variable with an exponential, and a modulus $2^{v+1} + 3$ that is not a prime power. We offer this as the most precise available *placement* of the difficulty; it is not a hardness theorem, and we make no claim that the difference is essential.

It is natural to read the impossibility results against Skolem’s conjecture (an unsolvable purely exponential equation is unsolvable modulo some witness [BH15]): the ten machine theorems are explicit local witnesses, and Theorem 5.5 says the flagships admit no witness in the congruence class. We stress that this is an orientation and not an application: the linear occurrences of k and of δv place our equations outside the purely exponential class the conjecture concerns. [INTERPRETATION]

11.1 Open problems

Question 11.2. Does the Space Needle halt—equivalently, does the orbit of 6 under (1) reach a power of two? The per-branch obstruction is exactly (2), and 120 census members share its sieve group, giving a graded family of test cases.

Question 11.3. Is there a 2-automatic invariant separating the Needle’s orbit from the powers of two, at any number of states? After Theorem 5.5 this is the minimal certificate class not proved empty.

Question 11.4. Is the one-schema family Turing-complete—some single member with undecidable halting set, or the parametrised problem? Remark 8.2 rules out the counting genre; the rigid slice of Theorem 8.1 is provably not where universality lives; universal generalised Collatz maps with a fixed modulus exist [Kaš92], and integer piecewise-affine reachability is delicate already in low dimension [BA13], but all known reductions spend arbitrarily many independent branch coefficients where this family has five.

Question 11.5. Prove that almost no start halts, with unbounded depth: for the Needle’s reduction, $\#\{x \leq N \text{ halting}\} = O(N^c)$ with $c < 1$. Depth-graded polylogarithmic counts are known exactly for each fixed depth, and measured branching along the backward tree peaks at $0.79 < 1$ [MEASURED], so a uniform subcriticality argument is plausible. This is the expanding-map mirror of backward-tree counting [KL03, KL10]; note that almost-everywhere results in the contracting regime [Tao22] are not transportable, the drift having the wrong sign.

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